

# RecruitConnect OneHR™

One Platform. Complete Workforce Intelligence.

Intelligent Workforce Operating System — not just HRIS

Version 1.0 • 2026-09-02 • MSSQL onehr\_v2 • RBAC hardened • 44 modules



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Tip: All pages linked in left Sidebar !' Manual. PDF also at </manual> and </public/manual.pdf>

## 1. Executive Summary

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OneHR answers “What is happening across your workforce right now?” — not just “Who are your employees?”. It merges People, Work, Time, Performance and Prediction into a single operating system. Four pillars guide the product:

- **Manage People:** hire !' alumni, digital identity RC-000245 + QR
- **Manage Work:** attendance, shifts (14 arrangements), tasks, rosters
- **Measure:** 6 health scores !' HR Health 89/100, activity `` productivity
- **Predict:** AI copilot (RAG), digital twin, simulator (what if salary +10%?)

Target: SMEs, corporates, government, schools, hospitals, banks, retail, manufacturing, NGOs. Industry templates (banking, school, hospital, manufacturing, retail, NGO, tech, generic) tune shifts, compliance and workflows.

## 2. Product Principles

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Activity `` Productivity — portal clicks never auto-score productivity. Supporting info only: login sessions, workflow actions, tasks completed, etc., plus KPI + manager assessment !' Overall Indicator.

- **Optional & Configurable:** Face/GPS/biometric opt-in per org, consent, 90-day retention, access-controlled.
- **Flag, Don't Accuse:** Fraud !' “Requires Review” (device sharing, duplicate face, impossible travel) never auto-accused.
- **Permission-Scoped AI:** Copilot answers only from approved policies, RLS per organization\_id.

### 3. User Roles & RBAC

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RBAC is dynamic. Super Admin creates custom roles (e.g., Finance Manager) and assigns permissions per module. Permissions are enforced in API (RbacGuard, RequirePermissions) and filtered in UI (Sidebar + page guards).

System roles fallback matrix (custom roles override via DB RoleDefinition.permissions JSON):

|                    |                                                                                        |
|--------------------|----------------------------------------------------------------------------------------|
| <b>super_admin</b> | * (all)                                                                                |
| <b>org_admin</b>   | * (all within org)                                                                     |
| <b>hr_admin</b>    | employee:*, attendance:*, leave:*, shift:*, payroll:*, job:*, learning:*, compliance:* |
| <b>manager</b>     | employee:read:team, attendance:read:team, leave:approve:team, task:*                   |
| <b>employee</b>    | self: read/update self, attendance:clock, leave:request:self, payroll:read             |
| <b>recruiter</b>   | job:*, vacancy:*, employee:read                                                        |
| <b>executive</b>   | report:read, analytics:read                                                            |

Login returns JWT with {sub, role, customRoleId, org\_id, permissions}. Sidebar filters NAV by role + permissions (hasPermission). Super Admin !' Admin !' Roles tab to create/edit, checkbox matrix grouped by module (44 modules).

### 4. Workforce Model (6 Layers)

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- 1 People — Digital Identity, skills, org structure
- 2 Work — tasks, projects
- 3 Time — sessions (Net = Gross " Breaks), overtime, late
- 4 Performance — KPI, appraisals
- 5 Engagement — surveys, life events
- 6 Intelligence — scores, risks, forecasts

## 5. Core Modules (44) — What’s Implemented Now

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All 24 dashboards are live (Settings, Projects, Workflows, Compliance, Engagement, Integrations, Documents, Performance, Assets, Shifts, AI Copilot, Onboarding, Leave, Analytics, Reports, Payroll, Recruitment, etc.). Each connects People!Work!Time!Performance.

- **People:** RC-000245, QR, 25+ fields
- **Attendance:** 7 methods, face 98%, motion 0.8–12%
- **Shifts:** fixed/flex/rotational/split/24h
- **Leave:** types, balances, workflow
- **Payroll:** OneHRCon merged, simulator
- **Recruitment:** ATS
- **Documents/Assets:**
- **Performance/PI:**
- **Engagement Surveys:**
- **Analytics:** HealthRadar, BranchBar
- **Reports:** DataGrid slicing/dicing
- **AI Copilot:** RAG
- **Workflows:** Automation
- **Integrations:** webhooks
- **Departments/Branches:** API-driven dropdowns

## 6. Smart Attendance & Face Verification

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Clock In/Out, Break Start/End !' WorkSession (clockInAt, breaks[], clockOutAt, gross/net/overtime, scheduled 480m). VerificationScore 98% for facial liveness. Methods: mobile, web, QR, biometric, facial, NFC, API. Snapshot retention 90 days.

- **Face Enrollment:** Employee receives link /face-enroll/:id (or People !' Camera icon !' Copy link). Captures 3 angles via camera, FaceDetector + motion 0.8–12% !' liveness verified. Stored as faceProfileRef JSON (8k truncated) + consentFace.
- **Clock-in Fraud Checks:** Device sharing (same fingerprint 7d), duplicate face (same snap 10m), liveness failed (motion out of range), no enrolled face, face mismatch (<60% !' critical duplicate\_face, <80% !' suspicious), no face snapshot. All flagged “Requires Review” in Exception Center.
- **Manual vs Auto:** Clock in/out is manual (button + face snap). Clock-out if forgotten: Superadmin determines time via Attendance !' Missing Clock-Outs (input time per session or Auto-close today at 17:00). Gross/Net recalculated, exception resolved.

## 7. People Bulk, Shifts, Leave, Payroll

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Bulk Upload (300+): People !' Bulk Upload (Excel) !' Download Excel Template with dropdowns (employment\_type, work\_arrangement, grade, department, branch linked). Hidden Dropdowns sheet + data validation for 300 rows, Instructions sheet. Upload .xlsx !' POST /employees/bulk/excel (multipart) or CSV !' POST /employees/bulk {csv}. Max 500/batch, 10mb. Auto-creates departments/branches if names not found, creates user Employee@123 if email provided.

- **People Edit:** Pencil !' modal (jobTitle, grade, department/branch linked via ?branchId, workArrangement, employmentType, status, skills, role). PATCH /employees/:id (RBAC: hr\_admin/org\_admin or self limited).
- **Shifts/Rosters:** GET/POST /shifts, POST /rosters (employee\_ids + dates).
- **Leave:** GET /leave/types !' POST /leave/requests !' PATCH /approve/reject (manager/team). Balances, calendar, notifications.

## 8. Command Centers

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HR Command Center (/hr): Live workforce (Employees 1245, Clocked In 1067, Exceptions 17, Live Map by branch, Attendance Trend). Manager (/manager): My Team Today/Month. Employee (/employee): Home + popup "Ready to clock in?" ! Attendance. Executive (/executive): HR Health 89/100, People Mix Donut, Risk, Simulator.

## 9. Workforce Intelligence & AI

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Workforce Score: 6 indicators !' HR Health. Risk Engine: expired certs, overdue reviews. Early Warning: lateness, absence. AI Copilot: 5 modes (HR/Manager/Executive/Recruiter/Employee), permission-scoped RAG over handbook Knowledge Vault. Digital Twin: simulate open 5 branches !' employees/skills/cost. Simulator: salary +10% !' payroll impact.



## 10. Architecture (MSSQL)

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Modular monolith NestJS 10 + Next.js 14 App Router + MSSQL onehr\_v2 (migrated from PostgreSQL). 46 tables, 22 modules. Prisma provider sqlserver, Json !' String @db.NVarChar(Max), String[] !' Json, enums !' String, NoAction cascades. Tenancy via organization\_id + sp\_set\_session\_context (was RLS).

- **DB:** MSSQL 2022 accountingappdb:1433, 46 tables, Prisma migrate dev, generate, seed (RC org, admin/superadmin, 3 plans, RBAC).
- **API:** NestJS, JWT 15m/7d, APP\_GUARD JwtAuthGuard + RbacGuard (dynamic DB permissions), ValidationPipe, CORS, rate limit 100/min, Swagger /api/docs, body 10mb for face base64.
- **Web:** Next.js 14 App Router, Tailwind, GlassCard/Pill, DataGrid (pagination 10/25/50/100, sorting, per-col filter, group-by, slice, pivot, CSV export), lucide-react, charts (HealthRadar, BranchBar).
- **Infra:** Redis (queues), S3 (snapshots/docs), BullMQ Automation Engine (Event!'Condition!'Action!'Notification!'Escalation), Meilisearch (search), Cron.

## 11. API Quick Start & Demo Accounts

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Base: <http://localhost:3001/v1>, Web: <http://localhost:3000>

# Health

curl <http://localhost:3001/v1/health>

# Login

curl -X POST <http://localhost:3001/v1/auth/login> -H "Content-Type: application/json" -d '{"email":"admin@recruitconnect.ng","password":"Admin@123"}'

# Create org (public)

curl -X POST <http://localhost:3001/v1/organizations> -d '{"name":"Acme Ltd","acronym":"ACM","industryTemplate":"tech","adminEmail":"acme@demo.ng","adminPassword":"Acme@123"}'

# Bulk template Excel

curl -H "Authorization: Bearer \$TOKEN" <http://localhost:3001/v1/employees/bulk/template/xlsx> -o template.xlsx

Demo: admin@recruitconnect.ng/Admin@123 (org\_admin), superadmin@recruitconnect.ng/Super@123, hr@/manager@/employee@ / Test@123, testco@demo.ng/Test@123

## 12. Operations Manual

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Landing !' Login (AuthNav shows Dashboard/Logout when authenticated) !' Role-based redirect. Employee sees popup "Go to Attendance" if not clocked today (sessionStorage). Attendance requires face + motion snap (98% if verified). HR/Admin resolves exceptions. Super Admin !' Admin !' Organizations tab shows all orgs (TC, RC) with Onboarding Queue (pending = no plan) !' Assign plan (POST /admin/subscriptions/assign). People !' Add/Bulk (Excel dropdowns) !' instantly visible in Organizations \_count.

- **Dropdowns:** All selects API-driven: GET /departments?branchId=, GET /branches, GET /admin/roles, GET /admin/permissions/grouped, GET /admin/modules. Linked: branch !' department. CRUD at Settings !' Dropdowns (Branches/Departments/Roles/Modules) via POST/PATCH/DELETE.
- **Reports:** DataGrid generic: pagination, sorting, per-col filter, global search, Group-by (count/sum), Slice (column+value), Pivot (row × col), Export CSV. Used in /reports and /analytics.
- **Face Enroll:** HR: People !' Camera icon !' Copy link /face-enroll/:id !' employee captures 3 snaps !' POST /employees/:id/face-profile. Clock-in compares live vs enrolled (<60% !' duplicate\_face critical).

## 13. Super Admin — Companies & Onboarding

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Register via /register !' POST /v1/organizations !' appears in Admin !' Organizations (pending, NEW badge, no plan). Superadmin assigns Starter/Growth/Enterprise (price, maxEmployees, modules) and optionally Company Group. Auto-creates Head Office branch + HR dept + leave types. Onboarding Queue shows action "Onboard !' Assign Plan". User sees only modules from assigned plan (checkModuleAccess).

## 14. Security & Privacy

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JWT 15m/7d httpOnly, argon2, RBAC, audit\_logs, TLS, AES at rest, S3 SSE, consentLog (face/gps), retention job deletes S3 + DB ref, right to erasure soft delete, device fingerprint hashed, location only if require\_gps && consent\_gps.

## 15. Roadmap & Support

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Phase 1-4 DONE (MVP !' Intelligence). Next: Extract Attendance service at 20k concurrent, Mobile offline queue, SSO (Azure AD), Biometric ZKTeco push. Support: hello@recruitconnect.ng, +234 800 123 4567, Victoria Island, Lagos. Docs: PRD, ERD, API\_SPEC, ARCHITECTURE, RBAC, ROADMAP in /docs.